UNIT 1 · KINEMATICS · WEEK 2

Proportionality & Linearization

Physics 2026–27 · assigned Thursday, October 8 · due Tuesday, October 13, before you graph the cart data

NAME _____ DATE _____

Why this matters. Next week the cart lab asks you to plot position x against t^2 — not against t. Before you can trust that move, you have to be able to look at an equation and *decide*, without guessing, which two variables will make a straight line. That decision is **linearization**, and it is the single skill behind every graph you will fit this year: carts, friction, energy, collisions, pendulums.

The vocabulary of proportionality

"IS PROPORTIONAL TO"

A variable y is **proportional to** x (written $y \propto x$) when one is a constant multiple of the other: $y = kx$, with k a constant. The test is simple: if you divide one by the other, the result never changes.

The exponent on a variable tells you how to *plot* it. The trick is to move every variable into the first power before you graph.

EQUATION	PROPORTIONALITY	PLOT THAT GIVES A STRAIGHT LINE	CURVED IF YOU PLOT…
$y = kx$	$y \propto x$	y vs x	—
$y = kx^2$	$y \propto x^2$	y vs x^2	y vs x
$y = k/x$	$y \propto 1/x$	y vs $1/x$	y vs x
$y^2 = kx$	$y \propto \sqrt{x}$	y^2 vs x	y vs x

ONE RULE, FOUR SHAPES: A STRAIGHT LINE APPEARS WHEN EACH SIDE OF THE EQUATION HOLDS A VARIABLE TO THE FIRST POWER.

The relationship to explore

THREE VARIABLES, ONE EQUATION

Three variables are tied together by

$$U^2 = 3P^2 / (\pi Q) \quad \text{where 3 and } \pi \text{ are constants.}$$

In the table, only one variable changes at a time; the other is held at 1, as the column headings note. Row $U = 2.0$ and row $U = 8.0$ are done for you. Fill in the rest. You may do this by hand, in Sheets/Excel, or any plotting tool.

U	U^2	Q $(P = 1)$	P $(Q = 1)$	$1/Q$ $(P = 1)$
1.0				
2.0	4.0	0.24	2.0	4.2
4.0				
8.0	64	0.015	8.2	67
12				
20				

READ EACH COLUMN'S HEADING BEFORE YOU COMPUTE IT — THE Q AND 1/Q COLUMNS HOLD P = 1; THE P COLUMN HOLDS Q = 1.

1.1 Plot it three ways

GRAPHING

Using your completed table, make **three** plots. "*something* vs *something*" always means **y (vertical) vs x (horizontal)**.

i. P vs U (with $Q = 1$)

ii. P vs U^2 (with $Q = 1$)

iii. $1/Q$ vs U^2 (with $P = 1$)

Desmos, Sheets, or hand-drawn graphs on graph paper all work. Label each axis with its variable and units where they apply.

1.2 Read the plots

PROPORTIONAL REASONING

a. Which of the three plots came out **linear** (straight)?

b. How could you have told *without graphing*? Answer in a few sentences and use the word **proportional**. (Hint: what power does each variable carry in $U^2 = 3P^2/(\pi Q)$?)

c. Now suppose U is held constant at $U = 1$. What plot of P and Q would give a straight line? Justify it with the equation.

YOUR REASONING

1.3 Bridge to the cart lab

WHY X VS T^2

A cart released from rest and rolling under constant acceleration obeys

$$x = \tfrac{1}{2}\, a\, t^2.$$

Next week's lab will ask you to plot x vs t^2, not x vs t. In two or three sentences, use the reasoning from this page to explain **why** x vs t^2 comes out straight — and what the **slope** of that line equals.

YOUR ANSWER

This is in-class practice (CW), not a problem set. Finish it before the cart lab so the x vs t^2 plot is a decision you make, not a rule you memorize.